KAGISO KOTU

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Professional Summary

Final-year BSc Computational and Applied Mathematics student at the University of the Witwatersrand, majoring in Computer Science and Applied Mathematics. Strong foundation in Java programming, Object-Oriented Programming, full-stack development, SQL and database design. Experienced in developing frontend interfaces, implementing application functionality and integrating backend services and databases through collaborative software projects. Analytical problem-solver with strong communication skills and a willingness to learn new technologies.

Education

University of the Witwatersrand, Johannesburg
BSc Computational and Applied Mathematics *Final-Year Student*
Majors: Computer Science and Applied Mathematics
Achievement: Invited member of the Golden Key International Honour Society

Technical Skills

Programming	Java, Python, JavaScript, C++, SQL
Frontend	HTML, CSS, JavaScript, Responsive Interfaces, Dashboard Development
Backend	Application Logic, REST APIs, Backend Integration, Data Handling
Databases	MySQL, MongoDB, Firebase, SQL, Database Design
Computer Science	Object-Oriented Programming, Data Structures and Algorithms
Tools	Git, GitHub, GitHub Actions, Android Studio
Practices	Agile Scrum, Test-Driven Development, CI/CD, Software Testing

Selected Technical Projects

Managrs – Freelancing Management Platform *JavaScript, HTML, CSS, MongoDB*

- Contributed to frontend development by building and improving the administration dashboard, user interfaces and interactive platform features.
- Worked on backend integration and database functionality using MongoDB to support application data and platform operations.
- Contributed to user management, filtering, reports, messaging and application management features.
- Collaborated with team members to connect frontend functionality with backend services and support a consistent user experience.

TNBT – Stokvel Management Platform *JavaScript, Firebase,CSS,React,Agile Scrum, TDD*

- Collaborated in a six-member team to develop a full-stack stokvel management application.
- Contributed to frontend interfaces and application functionality, focusing on how users interact with the platform.
- Worked on backend functionality and Firebase integration to support data storage and application operations.
- Participated in requirements engineering, software design, testing and deployment using Agile Scrum and GitHub Actions.

Woofer – Social Media Android Application *Java, MySQL, PHP REST APIs*

- Developed Android frontend features using Java, including user profiles, posts, comments, messaging and friend management.
- Integrated PHP REST APIs and MySQL database functionality to support backend operations and communication between the application and server.

Multiplayer Snake AI *Java*

- Developed a game agent using pathfinding and hazard-avoidance algorithms.
- Applied Object-Oriented Programming, algorithm design and logical problem-solving.

Experience

Applied Mathematics Tutor *University of the Witwatersrand*
Assist first-year students with mathematical concepts, analytical reasoning and problem-solving. Explain introductory Python programming concepts and help students apply computational techniques to problems.

Leadership & Community Involvement

High School Mentor: Provided younger learners with mentorship and academic guidance.
Community Volunteer: Participated in community outreach activities supporting residents at a local old-age home.

Core Strengths

Analytical Thinking • Problem Solving • Attention to Detail • Teamwork • Communication • Adaptability • Willingness to Learn